TABLE 1.—Units Applicable to the U.S. Standard Atmosphere 1976

	Symbol	Name	Quantity
Basic SI			
	m	meter	length
	kg	kilogram	mass
	s	second	time
	K	kelvin	thermodynamic temperature
	mol	mole	the amount of a substance
Derived SI			
	N	newton	force ($kg{\cdot}m/s^2$)
	Pa	pascal	pressure (N/m^2)
	J	joule	work, energy or quantity of heat ($N{\cdot}m$)
	W	watt	rate of energy (or heat) transfer (J/s)
Non-Standard			
	mb	millibar	pressure 100 (N/m^2)
	torr at 0°C	torr	pressure 133.322 (N/m^2)
	°C	Celsius degree	temperature kelvin minus 273.15
English			
	ft	foot	length 0.3048 m*

* exact definition

TABLE 2.—Adopted constants

Symbol	Value
A. Category I Constants	
k	1.380622×10^{-23} N·m/K
M_i	the set of the first 10 values (kg/kmol) listed in table 3
N_A	6.022169×10^{26} $kmol^{-1}$
R^*	8.31432×10^{-3} N·m/(kmol·K)
B. Category II Constants	
F_i	the set of the 10 values (dimensionless) listed in table 3
g_0	9.80665 m/s^2
g_0'	9.80665 $m^2/(s^2{\cdot}m')$
H_b	the set of eight values (km′) listed in table 4
$L_{M,b}$	the set of seven values (K/km′) listed in table 4
P_0	1.013250×10^5 N/m^2 (or Pa)
r_0	6.356766×10^3 km
T_0	288.15 K
S	110 K
β	1.458×10^{-6} $kg/(s{\cdot}m{\cdot}K^{1/2})$
γ	1.40 (dimensionless)
σ	3.65×10^{-10} m
C. Category III Constants	
a_i	the set of 5 values ($m^{-1}{\cdot}s^{-1}$) listed in table 6
b_i	the set of 5 values (dimensionless) listed in table 6
K_7	1.2×10^2 m^2/s
K_{10}	0.0 m^2/s
$L_{K,b}$	the set of 2 values (K/km) listed in table 5
$n(O)_7$	8.6×10^{16} m^{-3}
$n(H)_{11}$	8.0×10^{10} m^{-3}
q_i	the set of 4 values (km^{-3}) listed in table 7
Q_i	the set of 4 values (km^{-3}) listed in table 7
T_9	240.0 K
T_∞	1000.0 K
u_i	the set of 4 values (km) listed in table 7
U_i	the set of 4 values (km) listed in table 7
w_i	the set of 4 values (km^{-3}) listed in table 7
W_i	the set of 4 values (km^{-3}) listed in table 7
Z_b	the set of 6 values (km) listed in table 5
α_i	the set of 6 values (dimensionless) listed in table 6
ϕ	7.2×10^{11} $m^{-2}{\cdot}s^{-1}$

category I constants and a suitable set of equations, are sufficient to define that portion of the 1976 Standard Atmosphere below 86 km. This category includes nine single-valued and three multi-valued constants. Category III includes all the remaining constants which, along with category-I and category-II constants and the related equations plus an expansion of that set are necessary to define that portion of the 1976 Standard Atmosphere above 86 km. This category includes 7 single-valued and 11 multi-valued constants.

The constants, with appropriate dimensions and symbols, are listed according to categories in three successive sections of table 2.

The definition as well as the authority for the value of each constant is discussed separately from the tabular listing. The multi-valued constants, with one exception, have only their general symbol and dimensions listed in table 2, while the multiple values of these constants, i.e., one value for each of several gas species, or one value for each of several height levels, are listed in tables 4 through 7.

Discussion of the Adopted Values of the Primary Constants:

Category I Constants

k The Boltzmann constant, $k = 1.380622 \times 10^{-23}$ N·m/K, is theoretically equal to the ratio R^*/N_A, and has a value, consistent with the carbon-12 scale, as cited by Mechtly (1973).

M_i The set of values of molecular weights M_i listed in table 3 is based upon the carbon-12 isotope scale for which $C^{12} = 12$. This scale was adopted in 1961 at the Montreal meeting of the International Union of Pure and Applied Chemistry.

N_A The Avogadro constant, $N_A = 6.022169 \times 10^{26}$ $kmol^{-1}$, is consistent with the

1.3.11 DYNAMIC VISCOSITY.—The coefficient of dynamic viscosity μ ($N \cdot s/m^2$) is defined as a coefficient of internal friction developed where gas regions move adjacent to each other at different velocities. The following expression, basically from kinetic theory, but with constants derived from experiment, is used for computation of the tables:

$$\mu = \frac{\beta \cdot T^{3/2}}{T + S}. \qquad (51)$$

In this equation β is a constant equal to 1.458×10^{-6} kg/(s · m · $K^{1/2}$) and S is Sutherland's constant, equal to 110.4 K, both defined in table 2B. Because of the empirical nature of this equation, no attempt has been made to transform it into one involving T_M.

Eq (51) fails for conditions of very high and very low temperatures, and under conditions occurring at great altitudes. Consequently, tabular entries for coefficient of dynamic viscosity are terminated at 86 km. For these reasons caution is necessary in making measurements involving probes and other objects which are small with respect to the mean free path of molecules particularly in the region of 32 to 86 km.

The variation of dynamic viscosity with altitude is shown in figure 14.

1.3.12 KINEMATIC VISCOSITY.—Kinematic viscosity η is defined as the ratio of the dynamic viscosity of a gas to the density of that gas; that is,

$$\eta = \frac{\mu}{\rho}. \qquad (52)$$

Limitations of this equation are comparable to those associated with dynamic viscosity, and consequently tabular entries of kinematic viscosity are also terminated at the 86-km level. See figure 15 for a graphical representation of the variation of kinematic viscosity with altitude.

1.3.13 COEFFICIENT OF THERMAL CONDUCTIVITY.—The empirical expression adopted for purposes of developing tabular values of the coefficient of thermal conductivity k_t for heights up to the 86-km level is as follows:

$$k_t = \frac{2.64638 \times 10^{-3} \cdot T^{3/2}}{T + 245.4 \times 10^{-(12/T)}} \qquad (53)$$

This expression differs from that used in the U. S. Standard Atmosphere, 1962 in that the numerical constant has been adjusted to accommodate a conversion of the related energy unit from the temperature-dependent kilogram calorie to the invariant joule. Thus, the values of k_t in units of J/(m · s · K) or W/(m · K) are greater than the values of k_t in units of kcal/(m · s · K) by a factor of exactly 4.18580×10^3, when the kilocalorie is assumed to be the one for 15°C. Kinetic-theory de-

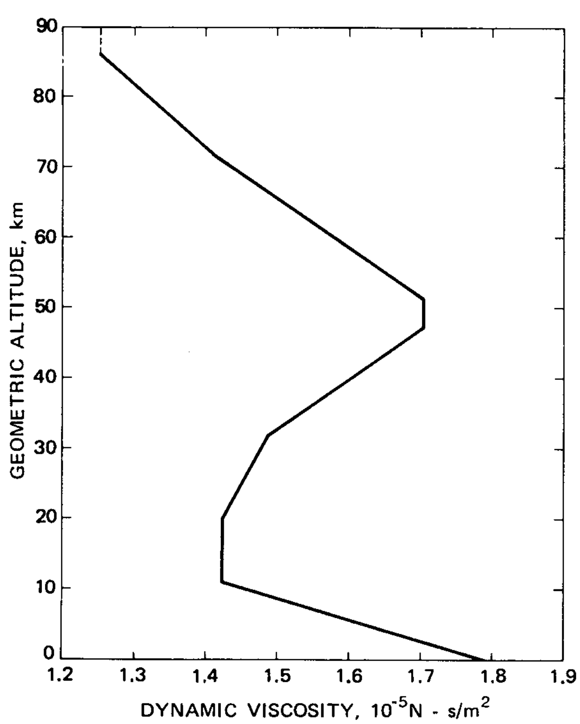

Figure 14. Dynamic viscosity as a function of geometric altitude

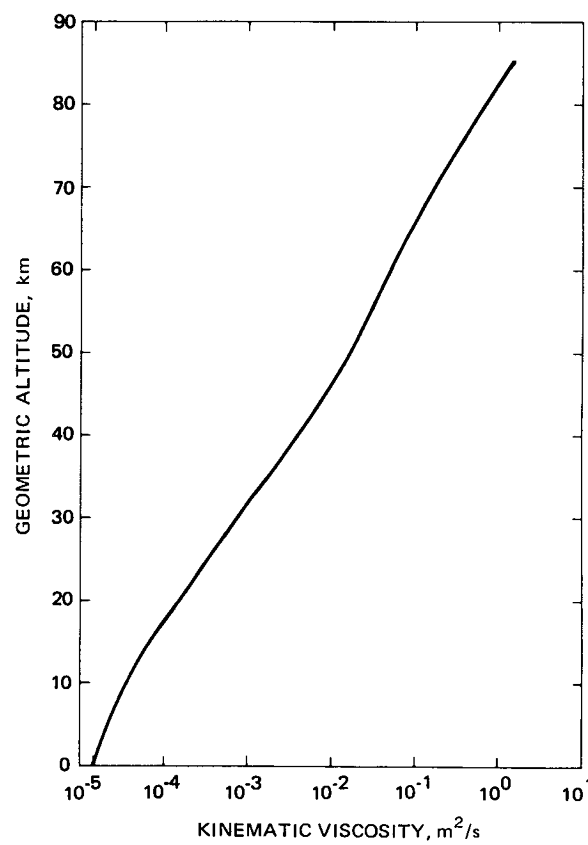

Figure 15. Kinematic viscosity as a function of geometric altitude